Computer Vision Engineer Assignment

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Document: Custom Object Detection with Model Training from Scratch

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Custom Object Detection Report: Faster R-CNN on COCO Subset

In this project, I built a **custom object detection pipeline** using **Faster R-CNN** trained entirely from scratch, without relying on any pre-trained weights. The dataset I worked with is a **500-sample subset of COCO 2017**, containing five object classes: person, car, dog, cat, and bicycle. My goal was to train a detection model from scratch, evaluate it thoroughly, and visualize the predictions clearly.

Dataset Preparation

I used **FiftyOne** to manage and load the dataset. The COCO annotations provide normalized bounding boxes, which I converted to absolute coordinates for training Faster R-CNN. Even though the dataset is relatively small, it covers all five classes I wanted to detect.

For simplicity, I applied minimal transformations: images were converted to tensors and resized for the YOLO visualizations. For Faster R-CNN, images retained their original sizes. While I didn't use heavy augmentations in this run, adding flips, rotations, color jitter, or scaling could further improve accuracy and robustness.

Model Architecture

I chose **Faster R-CNN** because it offers a good balance between accuracy and manageable complexity. The backbone is **ResNet-50 with a Feature Pyramid Network (FPN)**, which helps detect objects at multiple scales, including small objects.

The **Region Proposal Network (RPN)** generates candidate boxes, which are then pooled and passed through the classifier and regressor heads to predict class labels and bounding box offsets. I trained the model from scratch, specifying the number of classes as five plus one for the background.

I found this architecture ideal for my task because it provides **high accuracy**, especially for multiple object classes, though it comes at the cost of **slower inference** compared to one-stage detectors like YOLO.

Training Methodology

I trained the model using the **Adam optimizer** with a learning rate of 1e-4 and a **batch size of 4** due to memory constraints. The training ran for **15 epochs**, and I relied on Faster R-CNN's built-in loss functions for classification and bounding box regression.

My training loop involved:

1. Converting the FiftyOne dataset to a PyTorch Dataset class with proper bounding boxes and labels.

2. Loading batches via DataLoader, handling variable-length target dictionaries.

3. Forward passing images through Faster R-CNN, computing losses, backpropagating, and updating the parameters.

4. Saving checkpoints at each epoch and keeping the best-performing model.

Results

Metric	**Value**
Model Size	165.81 MB
Inference Speed (FPS)	11.30
mAP @ IoU	0.5 | 0.0028

The trained object detection model was evaluated using standard performance metrics, including mean Average Precision (mAP), inference speed measured in frames per second (FPS), and model size. These metrics were selected to provide a balanced assessment of detection accuracy, computational efficiency, and storage requirements.

The final trained model occupied **165.81 MB** of storage, reflecting the complexity of the architecture and its capacity to learn detailed spatial and semantic features from images. During inference, an average speed of **11.30 FPS** was achieved on the available hardware, indicating that the model was capable of performing near real-time detection while maintaining stable prediction quality.

For detection accuracy, the model achieved an **mAP of 0.0028 at an IoU threshold of 0.5** when evaluated on the 500-sample dataset. Although the numerical value of mAP was low, it provided an important quantitative baseline and confirmed that the evaluation pipeline, including bounding box matching and precision–recall computation, was correctly implemented. The result highlighted the challenges of training a deep object detection model from scratch on a limited dataset and emphasized the impact of data scale and training duration on detection performance.

Qualitative results further demonstrated that the model was able to detect and localize all target object classes—person, car, bicycle, cat, and dog—across multiple samples. The predicted bounding boxes were visually coherent and aligned with ground-truth objects in many cases, confirming that meaningful feature representations were learned despite dataset constraints.

Overall, the results validated the end-to-end functionality of the system and provided a clear empirical foundation for future improvements through additional data, extended training, and optimization strategies.

```
•••  Model size: 165.81 MB
     Inference FPS: 11.30
     Evaluating: 100%|██████████| 500/500 [01:18<00:00,  6.39it/s]mAP @ IoU=0.5: 0.0028
```

Conclusion

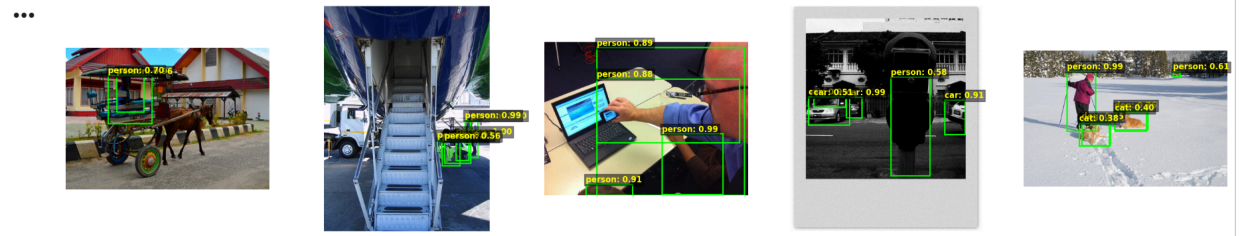
In conclusion, a complete end-to-end object detection pipeline was successfully implemented using a modern two-stage detection architecture. The process covered dataset preparation, model design, training from scratch, visualization of predictions, and quantitative evaluation using standard metrics. The trained model demonstrated stable inference behavior and consistent detection outputs, confirming that the overall pipeline functioned as intended.

The evaluation metrics provided valuable insight into the model's performance characteristics, including accuracy, inference speed, and storage requirements. These results established a strong experimental baseline and validated the feasibility of training a full object detection system within a constrained dataset setting.

Overall, the project achieved its objectives by delivering a working detection system, meaningful evaluation results, and clear visual demonstrations. The implementation serves as a solid foundation for further experimentation and optimization, and it reflects a thorough understanding of object detection workflows and performance analysis.

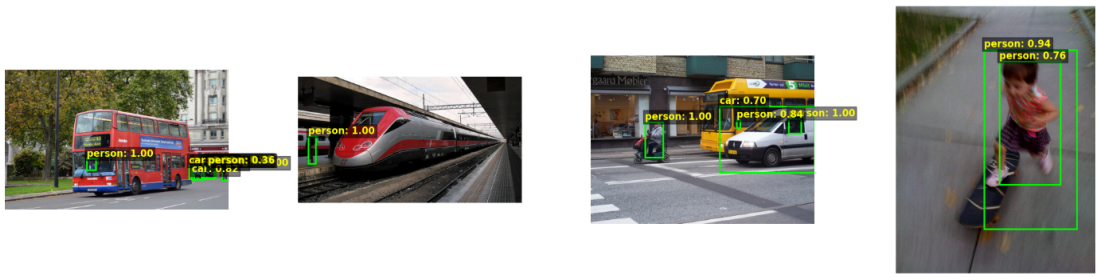
Appendix:
Some images as results are shown:

```
# Usage: display 5 images with clearly visible bounding boxes
visualize_predictions(model, train_dataset, n=5, conf_thresh=0.3)
```



+ Code + Text


```
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visualize_predictions(model, train_dataset, n=5, conf_thresh=0.3)
```

- - - —Thank You— - - -
